

Abstract

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1. RF Jammer

The purpose of this project is to generate and broadcast a high-amplitude RF signal (amp 50) containing Gaussian noise, which covers Wi-Fi and LTE signals. As we do not have the broadcasting device, we should simulate the broadcaster, the Wi-Fi receiver, and the LTE receiver in the GNU Radio and Python, if required. After that, we can analyze the Signal-to-Noise Ratio (SNR) in the conditions where the jamming signal is activated and when it is deactivated.

2. FM Radio

The purpose of this project is to process an FM radio signal by using GNU Radio. The project receives the raw signal, chooses the frequency by using Freq Xlating FIR filter block, and then decodes the FM signal to the audio signal by using WBFM Receive block. As the raw source input block is dependent on a physical device, the input must be simulated by preparing or generating a sample raw FM signal and developing a corresponding block with Python. If we decide to generate an FM signal, we can create a new project that converts the original wave source to the FM signal by reversing the process of the FM Radio receiver by using the FM modulator block.